

Supplementary Information for “Resistance-Aware Computational Analysis of Antimalarial Leads from African Natural Products”

August 8, 2026

Table 1: Docking protocol validation: enrichment metrics and re-docking accuracy for the four wild-type targets. Values reproduce the corrected benchmarks from the parent study (V2-corrected grids; AutoDock Vina + DiffDock consensus). MMV Malaria Box values are consensus-score ROC-AUCs; DEKOIS 2.0 validation used AutoDock Vina only. BEDROC values use $\alpha = 20$ (MMV). Re-docking RMSD is the heavy-atom RMSD between the top-scoring docked pose and the co-crystallised ligand; MTX redocking failed (RMSD ≈ 30 Å).

Target	Method	ROC-AUC	EF1 %	EF5 %	BEDROC
PfDHFR (7F3Y)	Vina redock	—	—	—	—
	MMV consensus (Vina+DiffDock)	0.924	—	2.57	1.309
	DEKOIS (Vina only)	0.509	—	0.51	—
PfCRT (6UKJ)	MMV consensus (Vina+DiffDock)	0.971	—	1.11	9.434
PfATP4 (9N10)	MMV consensus (Vina+DiffDock)	1.000	—	2.00	1.810
PfClpR (4GM2; not PfClpP)	MMV consensus (Vina+DiffDock)	1.000	—	2.00	1.810

Identity note.—PDB 4GM2 is PfClpR rather than PfClpP; the historical 4GM2 docking record is therefore not PfClpP validation.

Note.—Redocking against V2-corrected grids gave 100% success for the five alignable co-crystallised ligands. MTX (PfDHFR co-crystallised ligand) could not be redocked and is reported as N/A. AutoDock Vina + Gnina consensus scoring was used in the present validation study; the enrichment metrics above are taken from the parent study (Vina + DiffDock) to ensure consistency with the corrected P1 data.

Table 2: Endpoint-analysis status for the four parent-study consensus complexes used for production MD. These systems are distinct from the 17 set-C polypharmacology candidates. Mean $\pm$ standard deviation is shown only for the single interpretable MM-GBSA result from 101 snapshots; dissociated or conversion-corrupted systems are reported as N/A. Mutant RRS is a docking-based analysis of set C and is not assigned to these MD systems.

Ligand	Target	ΔG_{bind}	Std. dev.	Status
164	Historical PfClpP cohort label*-WT	—	—	N/A; ligand dissociated [‡]
201	PfDHFR-WT	—	—	N/A; ligand dissociated [‡]
214	PfCRT-WT	−18.25	0.40	Interpretable (gmx_MMPBSA)
438	PfATP4-WT	—	—	N/A; conversion corrupted [§]

*The 164–PfClpP label is historical; PDB 4GM2 is PfClpR rather than PfClpP and this system is not PfClpP

[‡]Ligand dissociated during production; no binding free energy is inferred.

[§]Excluded due to two-chain topology incompatibility with CHARMM36-to-AMBER conversion.

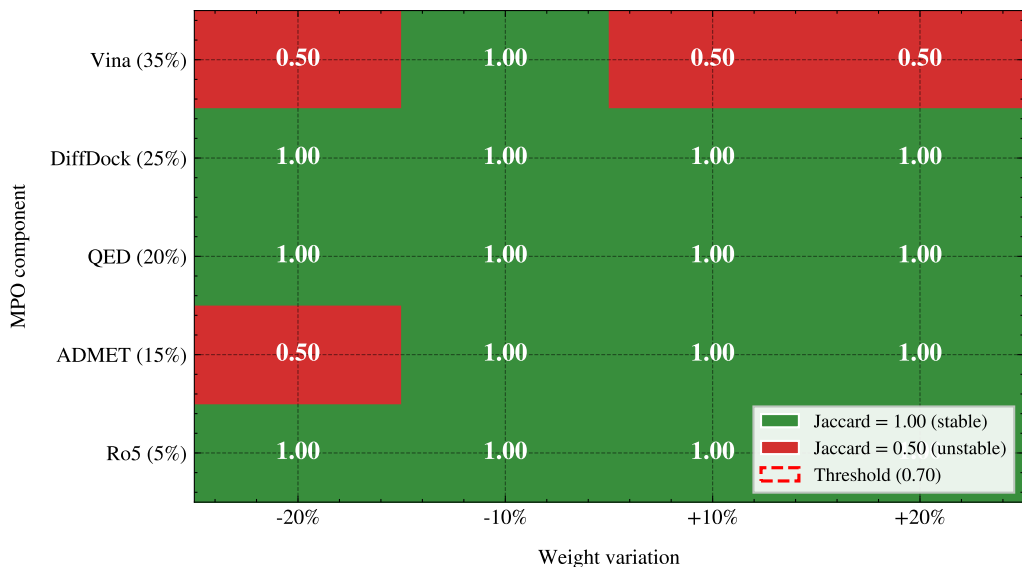


Figure 1: MPO sensitivity analysis heatmap. Each cell shows the Jaccard similarity of the top-20 hit list when the corresponding MPO weight is varied by $\pm 20\%$. Green (1.00) indicates the hit list is unchanged; red (0.50) indicates substantial reordering.

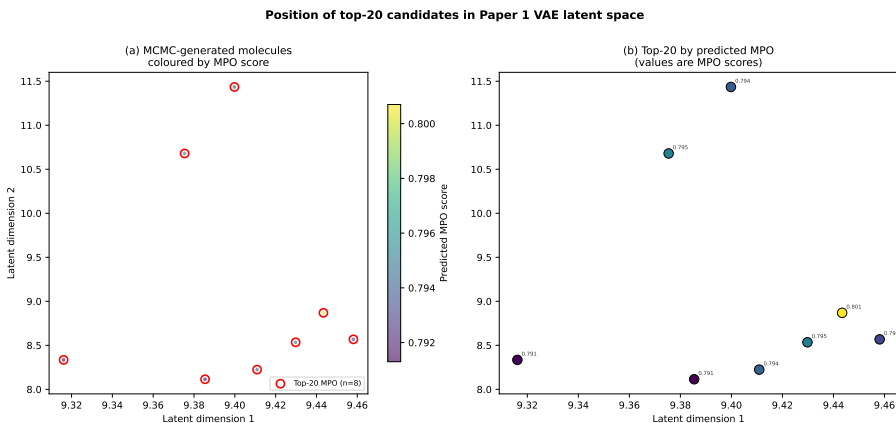


Figure 2: Position of the top-20 candidates in the Paper 1 VAE latent space. Molecules are coloured by predicted MPO score; top-20 by MPO are highlighted in red.

Table 3: Planned ADMET cross-validation schema for the 17 set-C candidates and 5 reference controls. The table is intentionally unpopulated in the canonical evidence set; the proposed comparison would use ADMET-AI, ADMETlab 3.0, and SwissADME, with Spearman correlations and inter-tool disagreement flags reported after completion.

Compound	ADMET-AI LogS	ADMETlab LogS	SwissADME LogS	hERG pIC ₅₀	Flagged?
<i>No completed three-tool ADMET comparison is included in the canonical evidence set.</i>					

Table 4: ACSI weight sensitivity analysis. Each of the four ACSI weights (w_{DrugBank} , w_{ANPDB} , w_{fsp} , w_{NPL}) was varied by $\pm 20\%$ while keeping the remaining weights proportional. The Jaccard index measures overlap of the resulting top-20 ACSI-ranked list with the baseline ranking. A Jaccard ≥ 0.70 (14/20 compounds stable) indicates robustness.

Weight varied	Variation (%)	Mean ACSI (top-20)	Std. dev.	Jaccard vs. baseline
<i>Data to be populated after ACSI computation.</i>				

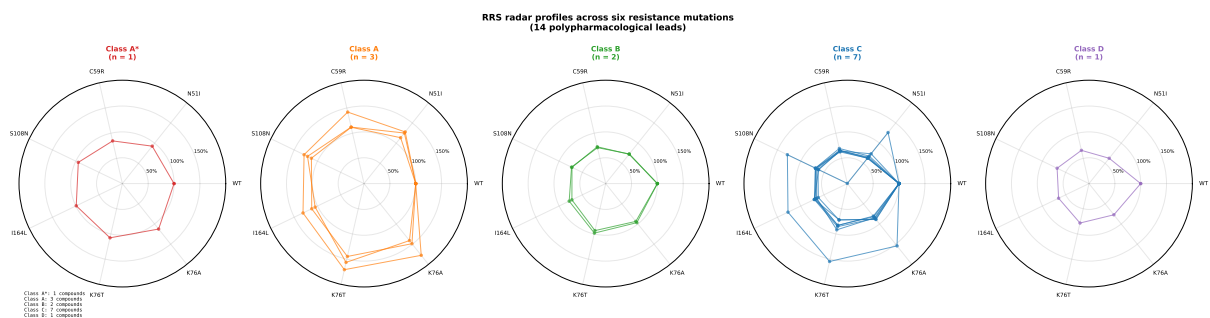


Figure 3: Per-compound, docking-derived RRS profiles across all six resistance mutations for the 17 set-C polypharmacological leads. The four parent-study MD systems are not included. Each axis represents one mutation; concentric rings indicate 50 %, 100 %, and 150 % RRS (WT-normalised). Class A* and Class A compounds fill the outer region (RRS > 100 %).